

1. Problem Definition

Antisemitism detection in social media text is more challenging than general hate speech classification. The boundary between criticism of Israeli policy and hate speech remains contested, particularly under the IHRA Working Definition. In addition, antisemitic discourse often relies on coded language, making identity-related terms alone unreliable indicators. Our goal was therefore to develop an effective antisemitism detection model guided by three main objectives:

- Strong positive accuracy,
- Relative independence from buzzwords and keywords,
- Robustness across other datasets.

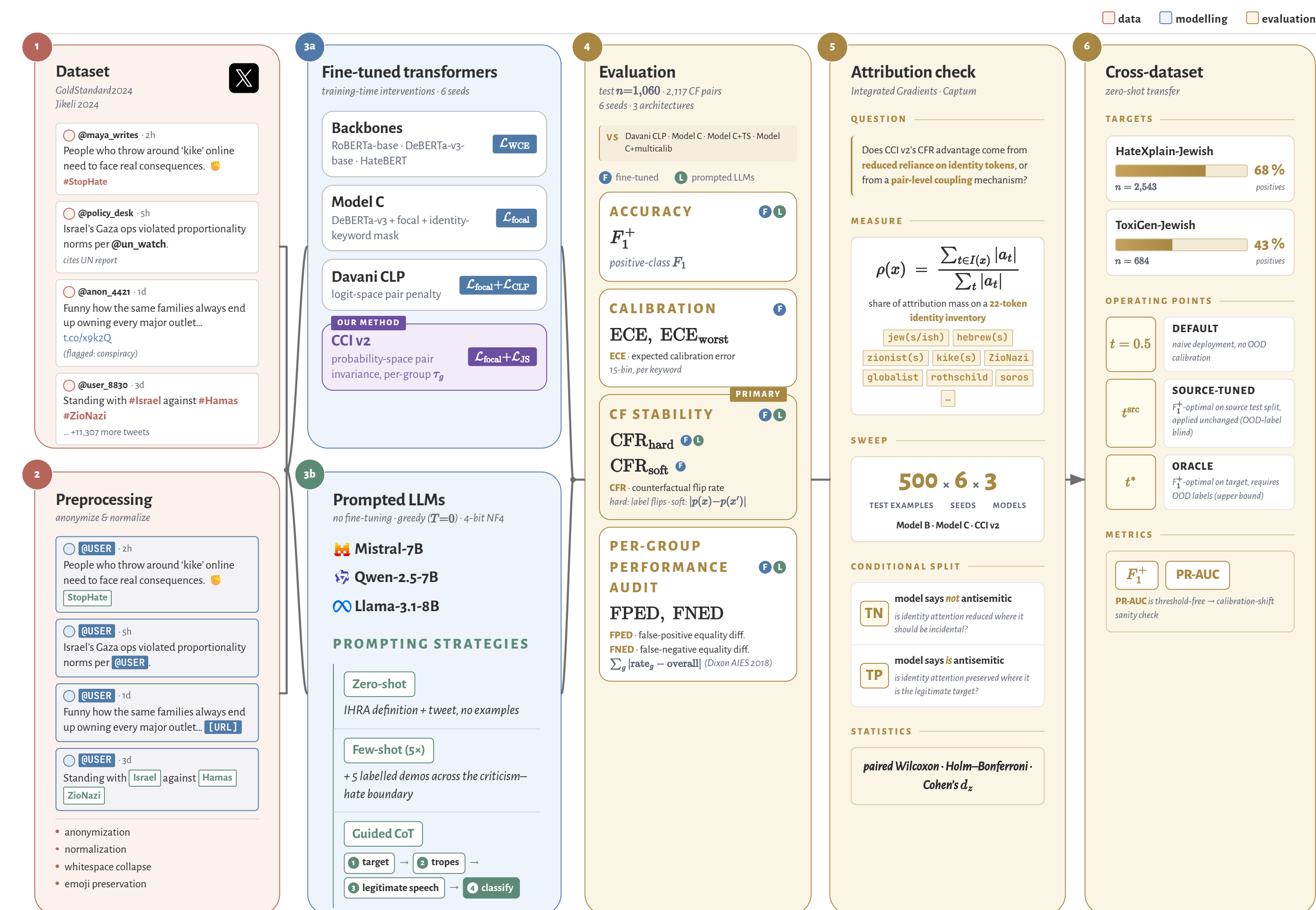
Our Contribution.

- A training-time loss, **Counterfactual Calibration Invariance (CCI v2)**, that penalises JS divergence between original and identity-swapped probability predictions.

2. Related Work

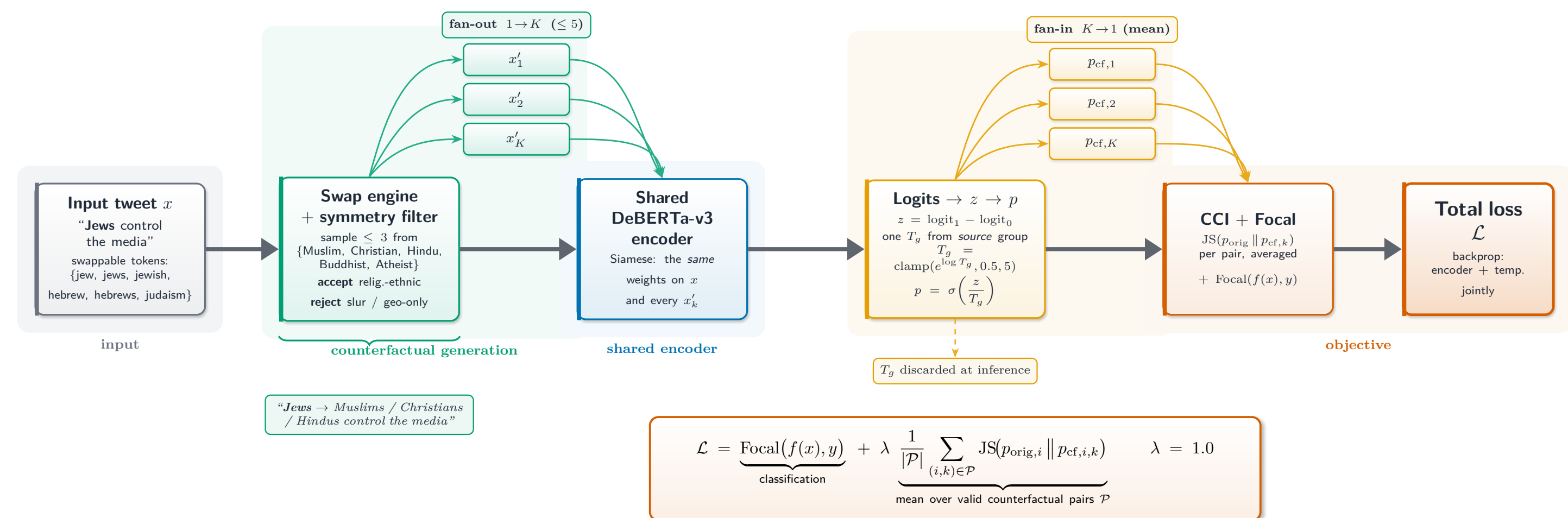
- Benchmark.** Jikeli et al. [1] introduce GoldStandard2024, an IHRA-annotated Twitter/X benchmark with strong label imbalance and keyword-dependent base rates.
- Counterfactual fairness.** Davani et al. [2] regularise paired counterfactuals in logit space; CCI v2 instead aligns probability distributions with a JS penalty.
- Calibration.** Temperature Scaling and Multicalibration [3] improve confidence estimates, but do not directly optimise counterfactual decision stability.

3. Method



CCI v2: Counterfactual Calibration Invariance

A training-time consistency loss with a shared (Siamese) encoder



Counterfactual test manifest → CFR

Fate of each source group on the 1,060-tweet test split. A pair (x, x') = one tweet x + one label-preserving identity swap x' ; groups bucketted by primary keyword (slur > proxy > religious). The resulting pair set P is the evaluation set for the counterfactual flip rate.

SOURCE GROUP	SYMMETRY-FILTER VERDICT	PAIRS IN MANIFEST P
Jews 598 tweets	religious anchor present ✓ rule 3 - accept	1,725
Israel 390 tweets	proxy and Jewish mention ✓ rule 3 - accept	392
$ P = 2,117$ symmetric pairs		

P evaluated by the Counterfactual Flip Rate (CFR), aggregated per pair	
HARD CFR (DECISION FLIPS)	SOFT CFR (PROBABILITY SHIFT)
$CFR_{hard} = \frac{1}{ P } \sum_{(x,x') \in P} \mathbb{1}[\hat{y}(x) \neq \hat{y}(x')]$	$CFR_{soft} = \frac{1}{ P } \sum_{(x,x') \in P} p(x) - p(x') $

Counterfactual test manifest.

Generated by the symmetry-filtered swap engine: **2,117 pairs** (Jews group 1,725; Israel group 392; Kikes and ZioNazi groups 0 by construction, since swaps would not preserve the label).

4. Datasets

GoldStandard2024 [1] comprises 11,311 IHRA-annotated English tweets (4.79:1 imbalance). Joint stratification on label and keyword yields:

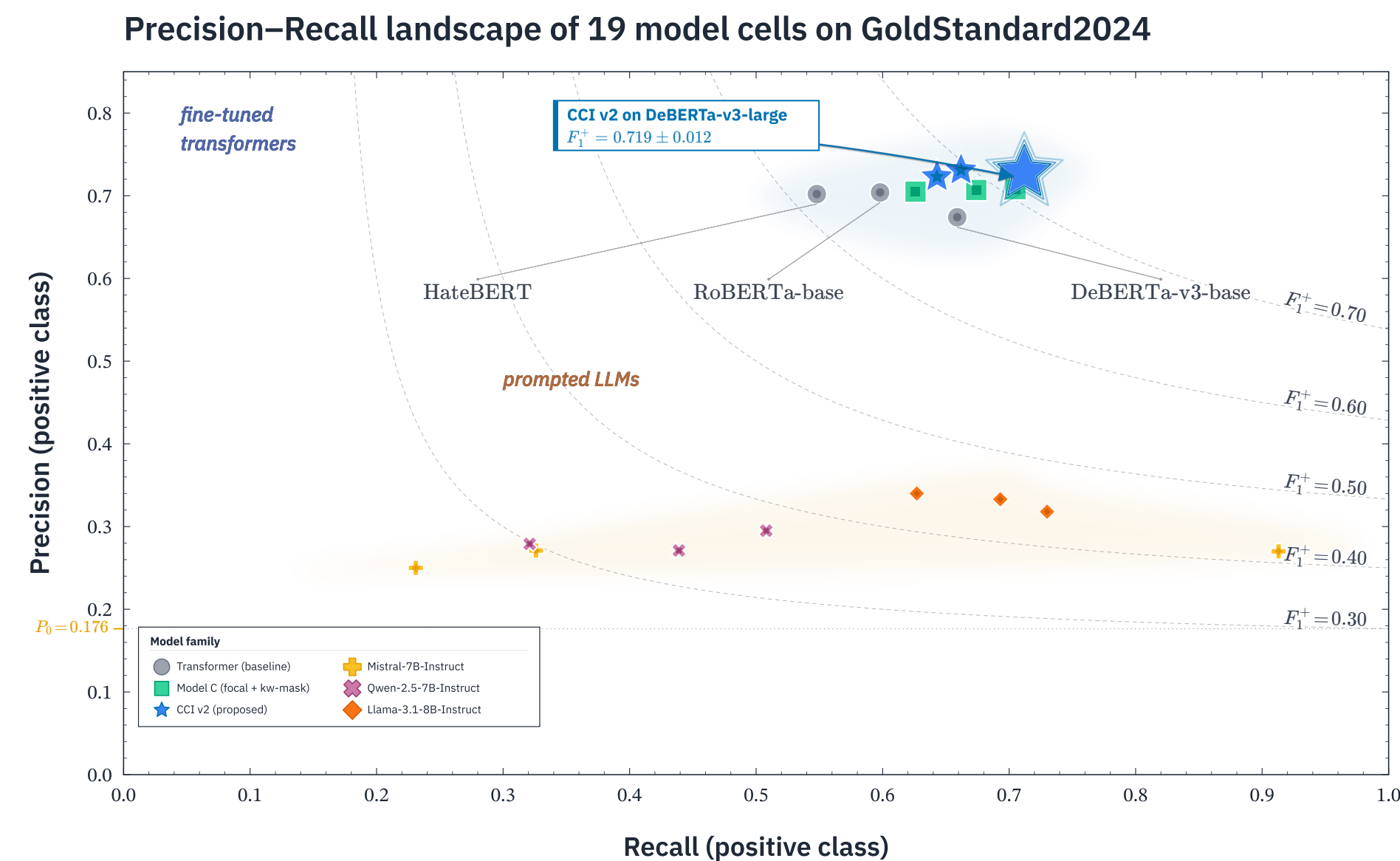
Keyword	Tweets	Anti.	Base rate	Split	Tweets
Jews	6365	725	11.4%	Train	8479
Israel	4134	664	16.1%	Dev	1060
ZioNazi	529	467	88.3%	Test	1060
Kikes	283	97	34.3%		
Total	11 311	1953	17.3%		

Test split contains 187 antisemitic tweets. All multi-seed numbers in this work are reported on this 1,060-tweet test set.

Generalization beyond the training distribution: we also evaluate the models on two external Jewish-targeted test sets: HateXplain Jewish, composed of 2,543 human-annotated posts, and ToxiGen Jewish, containing 684 machine-generated posts with toxicity annotations.

5. Results

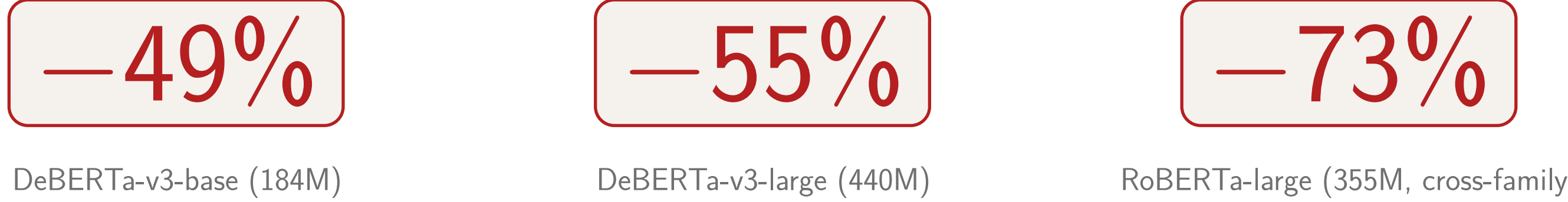
Result 1 Fine-tuned transformers outperform prompted LLMs. Prompted LLMs improve with stronger prompting, but they do not match fine-tuned transformers on this benchmark.



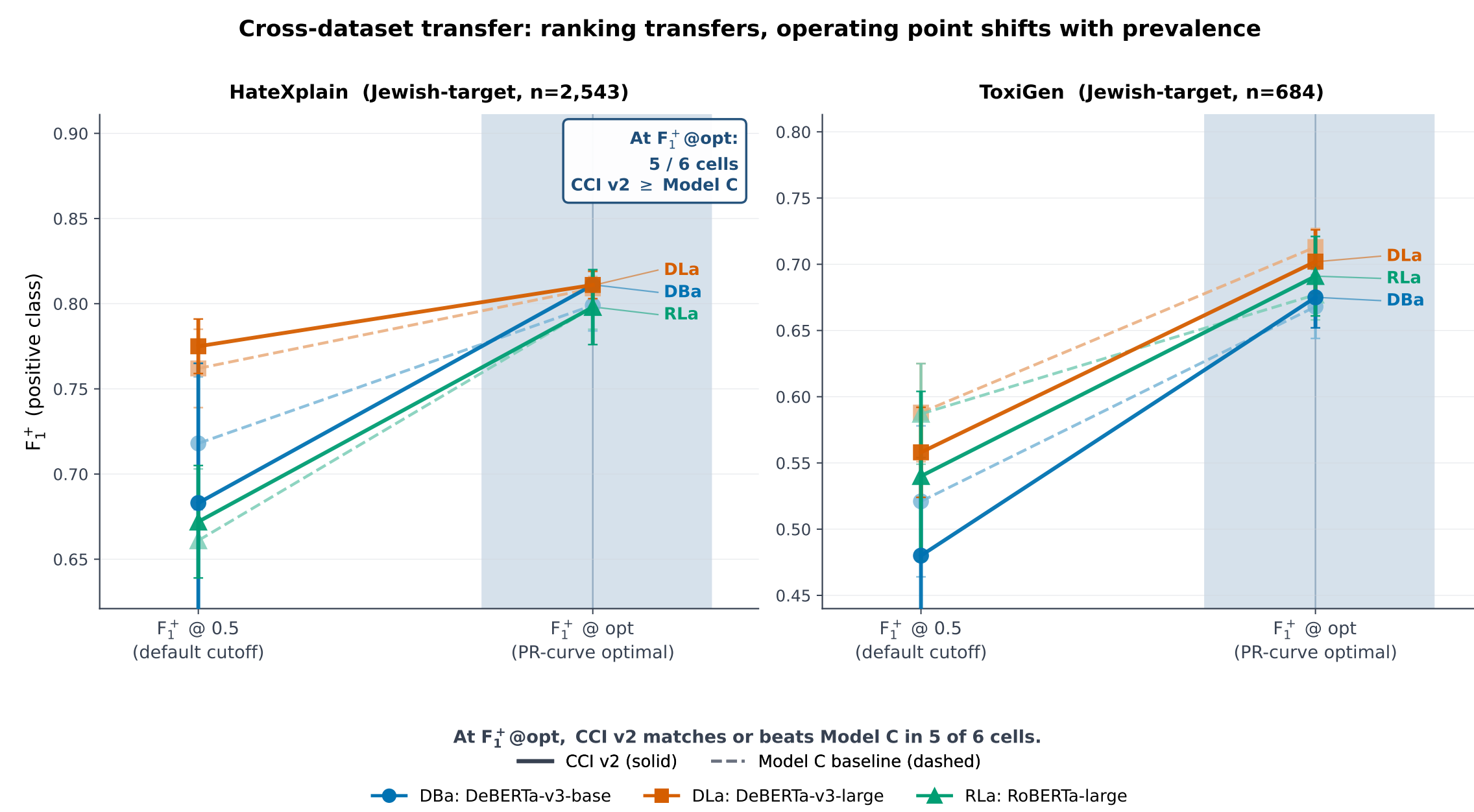
Result 2 CCI v2 leads the robustness-aware comparison. Across supervised models, CCI v2 delivers the strongest balance between positive accuracy and counterfactual robustness.

Method	F_1^+	ECE	CFR _{hard}	FNED
Model C (focal + kw-mask)	0.660	0.077	0.0358	0.584
Davani CLP, $\lambda = 0.5$	0.672	0.094	0.0403	0.528
CCI v2 (fixed T , JS) – pivot	0.677	0.098	0.0183	0.543
CCI v2 on DeBERTa-v3-large	0.719	0.099	0.0187	0.429
CCI v2 on RoBERTa-large	0.694	0.077	0.0126	0.560

Result 3 CCI v2 is the only intervention that consistently improves counterfactual robustness. CCI v2 cuts hard counterfactual flips across all three architectures, with large gains relative to the matching baseline.



Result 4 Cross-dataset transfer preserves ranking quality. Out of distribution, CCI v2 is more conservative at the default threshold and therefore misses some positives. After threshold tuning, however, it matches or outperforms the baseline in nearly every architecture–dataset setting.



6. Limitations

- Scope.** Training uses one English Twitter/X dataset; multilingual, cross-platform and longer-form moderation settings remain untested.
- Labels.** The target follows IHRA annotations, so ambiguous criticism–hate boundary cases depend on this annotation framework.
- Counterfactual coverage.** The symmetry filter protects label validity, but excludes slur/proxy groups such as Kikes and ZioNazi from the CFR manifest.
- Deployment.** OOD experiments reveal threshold shift: CCI v2 preserves ranking quality, but a fixed 0.5 cutoff may underpredict positives under prevalence shift.
- Granularity.** Binary labels collapse trope type, severity, sarcasm and counterspeech, so the model is better viewed as a triage aid.

7. Conclusion & Future Work

Takeaway. CCI v2 turns counterfactual fairness from a post-hoc audit into an explicit training signal: a tweet and a label-preserving identity swap should receive the same calibrated prediction. This reduces CFR_{hard} by 49–73% across DeBERTa and RoBERTa backbones, while maintaining strong F_1^+ and comparable FNED to the strongest baselines.

Mechanism. The gain is not explained only by stronger identity-token masking. The attribution audit shows only a small Model C–CCI difference on identity-token attribution, so the dominant effect is the learned pair-level coupling.

Next steps.

- Replace the rule-based symmetry filter with the LM-likelihood scorer.
- Add OOD threshold calibration for prevalence-shifted deployments.
- Test cross-language and cross-identity transfer.
- Move from binary detection to trope/severity-aware labels.

8. Key References

- G. Jikeli et al., “Antisemitism on Twitter: A dataset for machine learning,” Zenodo, v3, 2024, doi:10.5281/zenodo.14448399.
- A. M. Davani et al., “Improving counterfactual generation for fair hate speech detection,” WOAII, 2021.
- U. Hébert-Johnson et al., “Multicalibration: Calibration for the (computationally identifiable) masses,” ICML, 2018.